SPiiPlus MMI Application Studio

Quick Start Guide

June 2025

Document Revision: 4.10

MOTION CONTROL

TRADEMARKS

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NOTICE

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Revision History

Date	Revision	Description
Jun 2025	4.10	Updated screenshots to latest MMI version
Jan 2025	4.00	Updated formatting
Jul 2024	3.14b	Updated formatting
Feb 2024	3.14	New Release
Nov 2023	1.00	Initial Version

Conventions

We use the following conventions in this document.

Text Formats

Format	Description
Bold	Names of UI objects or commands
BOLD + UPPERCASE	ACSPL+ variables and commands
`Monospace + gray background`	Code example
Italics	Names of other documents
Blue	Hyperlink
[ ]	In commands indicates optional item(s)
|	In commands indicates either/or items

Flagged Text

	Note - includes more information or programming tips.
	Caution - describes a condition that may result in damage to equipment.
	Warning - describes a condition that may result in serious bodily injury or death.

Table of Contents

List of Figures

1. Introduction

This Quick Start Guide (QSG) describes how to set up and use the SPiiPlus MMI Application Studio to configure ACS products.

The ACS website provides documentation and videos that describe in detail the SPiiPlus MMI Application Studio and ACS hardware. Use the following link to access this material:

https://www.acsmotioncontrol.com/resources/

See More Information for additional resources.

This document assumes the host computer uses Windows 10. The SPiiPlus MMI Application Studio is compatible with other operating systems. Consult the Resources page of the ACS website for a list of compatible operating systems.

This document assumes that the controller you are working with is new or set to the factory default settings.

For release notes for all versions of the SPiiPlus MMI Application Studio, go to the ACS Motion Control website, then navigate to:
Downloads > **SPiiPlus MMI Application Studio** > **Download Resources**.

2. Download and Install the SPiiPlus Application Development Kit (ADK) Suite

The latest version of the SPiiPlus ADK Suite is available as a download from the ACS Motion Control website. The SPiiPlus ADK Suite includes the SPiiPlus MMI, SPiiPlus Simulator, drivers, programming examples, and documentation for ACS products.

Before attempting to download and install the SPiiPlus ADK Suite, note:

- **You must have an account with ACS Motion Control to download the SPiiPlus ADK Suite.** If you do not have an account, you will be prompted to request an account on the login page of the content you wish to download or view.
- **Host computer compatibility:** Before installing the SPiiPlus ADK Suite, verify that the host computer has a compatible operating system. See the SPiiPlus ADK Suite Release Notes (on the same web page as the downloadable SPiiPlus ADK Suite) to determine if your planned host computer has a compatible operating system.

Download and install the SPiiPlus ADK Suite:

1. On the Web, navigate to: https://www.acsmotioncontrol.com/resources/.
2. At the top of the page, select **Resources** > **Downloads** > **SPiiPlus ADK Suite**.
3. Select the SPiiPlus ADK Suite Zip file (see Figure 2-1). Click **Download**.
4. After the suite has downloaded, unzip the file and run the Installer.

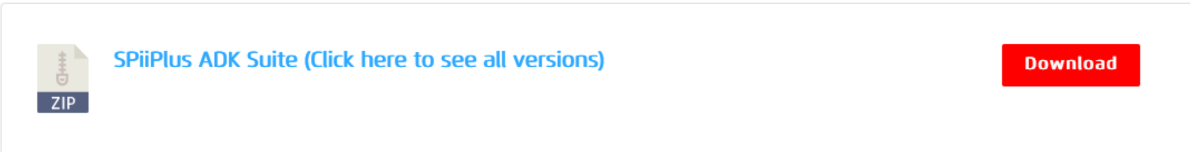

Figure 2-1. SPiiPlus ADK Suite on Resources Page of the ACS Website

3. Set up the Host Computer Ethernet Port

> If necessary, consult your IT department for assistance with the following steps for setting up an Ethernet connection on the host computer.

To setup the host computer Ethernet port:

1. In the Windows search bar, type **View Network Connections**, and open the control panel window when prompted (see Figure 3-1).
2. Within the **Network Connections** window, select the Ethernet connection that the controller will connect to.

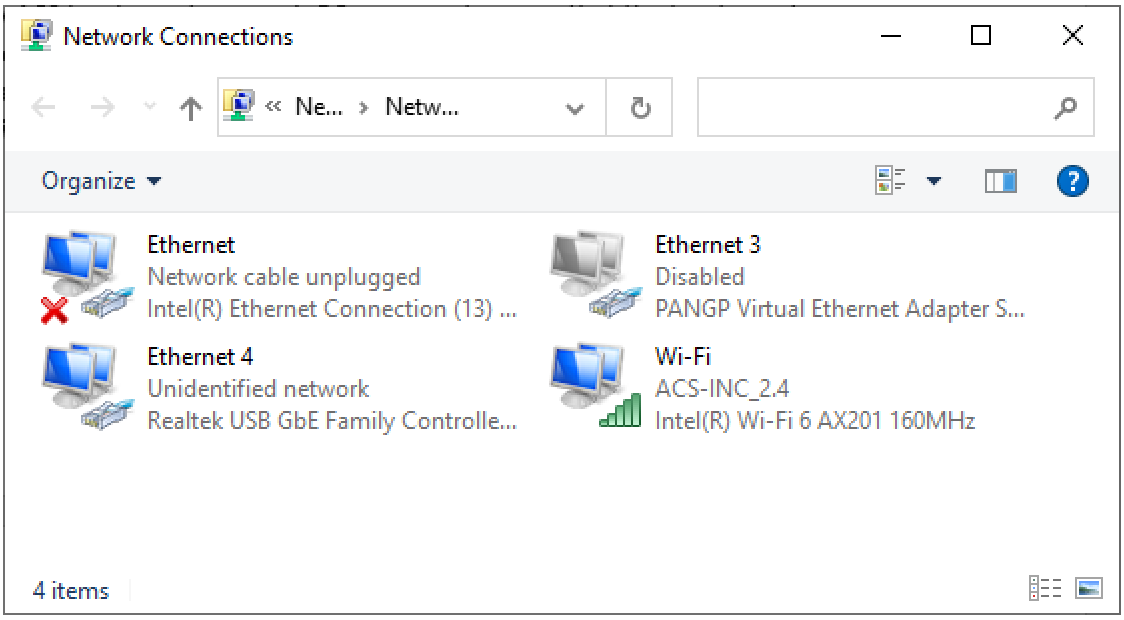

Figure 3-1. Network Connections Window

3. Right-click on the selected Ethernet port, then select **Properties** (see Figure 3-2).

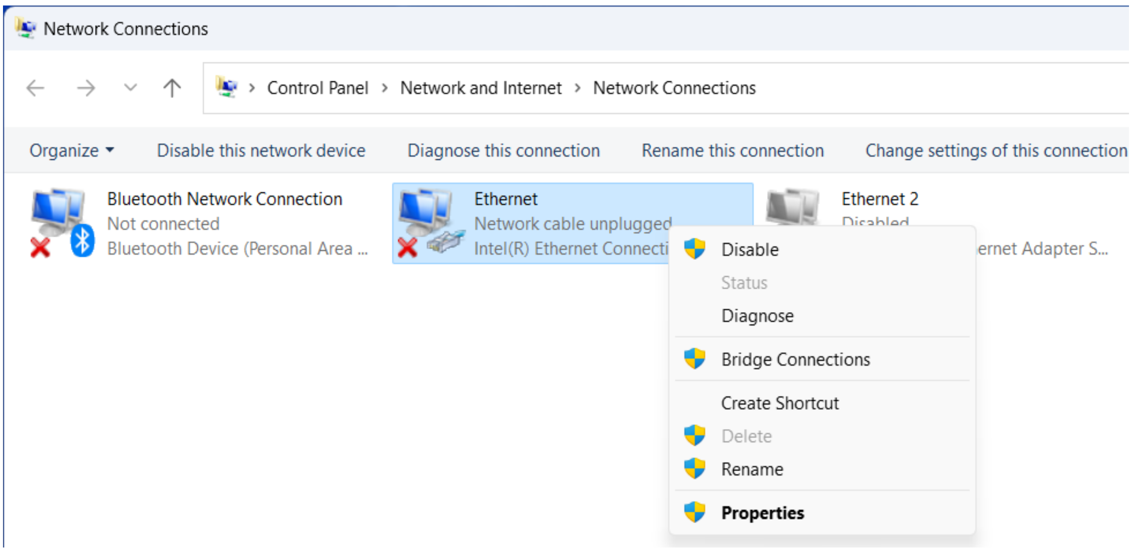

Figure 3-2. Select Properties

4. In the Ethernet Properties window, scroll down and select **Internet Protocol Version 4 (TCP/IPv4)**.
5. Select **Properties** (see Figure 3-3). This opens the **Internet Protocol Version 4 (TCP / IPv4) Properties** window (see Figure 3-4).

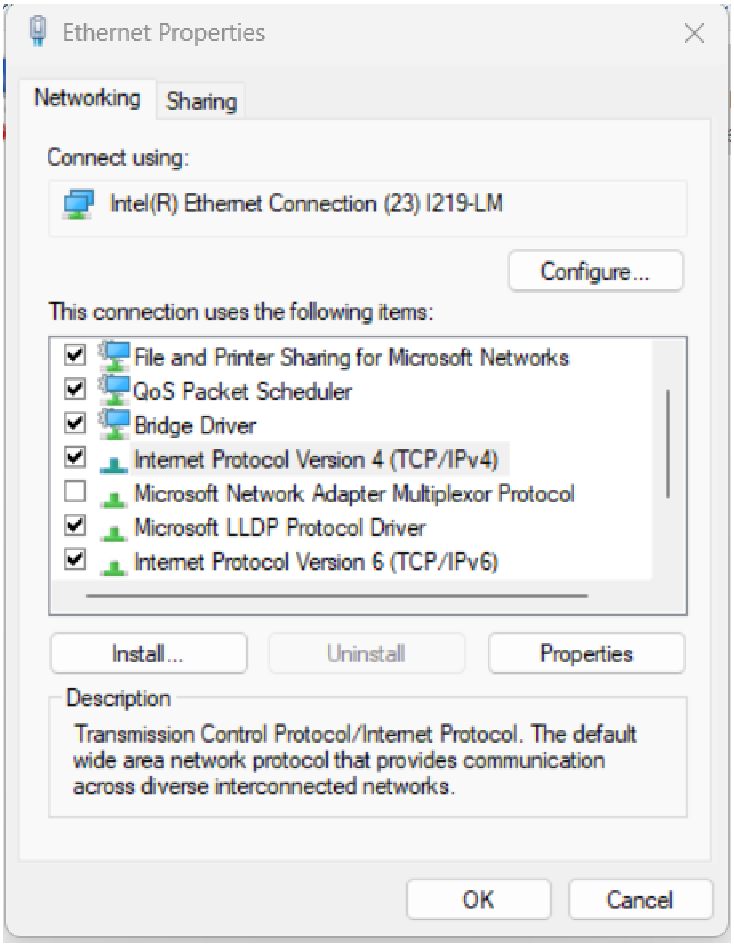

Figure 3-3. Ethernet Properties

6. Enter the following settings in the **Internet Protocol Version 4 (TCP / IPv4) Properties** window (see Figure 3-4):
   - Select **Use the following IP address**.
   - **IP address: 10.0.0.XYZ** (XYZ can be any number except 100, which is the default address for ACS controllers).
   - **Subnet mask: 255.255.255.0**.

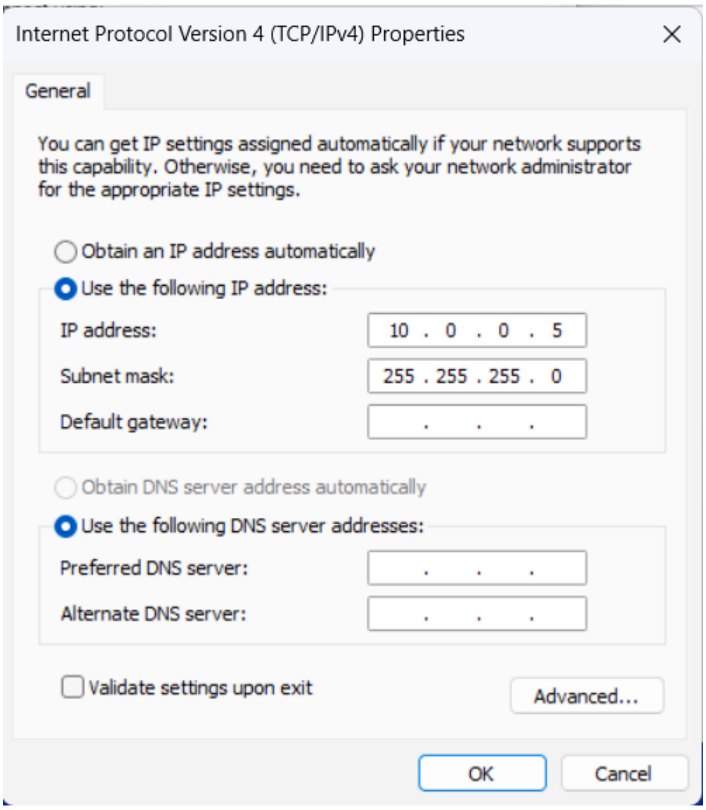

Figure 3-4. TCP/IPv4 Settings

7. Select **OK**, then close the **Network Settings** window.

4. Connect SPiiPlus MMI Application Studio to the Controller

Consult the hardware product guides before performing the following procedure. Product guides are available in:

- The ADK Suite under C:\Program Files (x86)\ACS Motion Control\SPiiPlus Documentation Kit\Product Guides.
- The ACS resource Library: https://www.acsmotioncontrol.com/resources/

To connect SPiiPlus MMI Application Studio to the controller:

1. Open the SPiiPlus MMI Application on the host computer.
2. Power up the hardware, then connect the controller to the Ethernet port you set up in the previous procedure.
3. In the **Workspace** window, right-click on **My Controller (Simulator)**, then select **Remove** (see Figure 4-1).

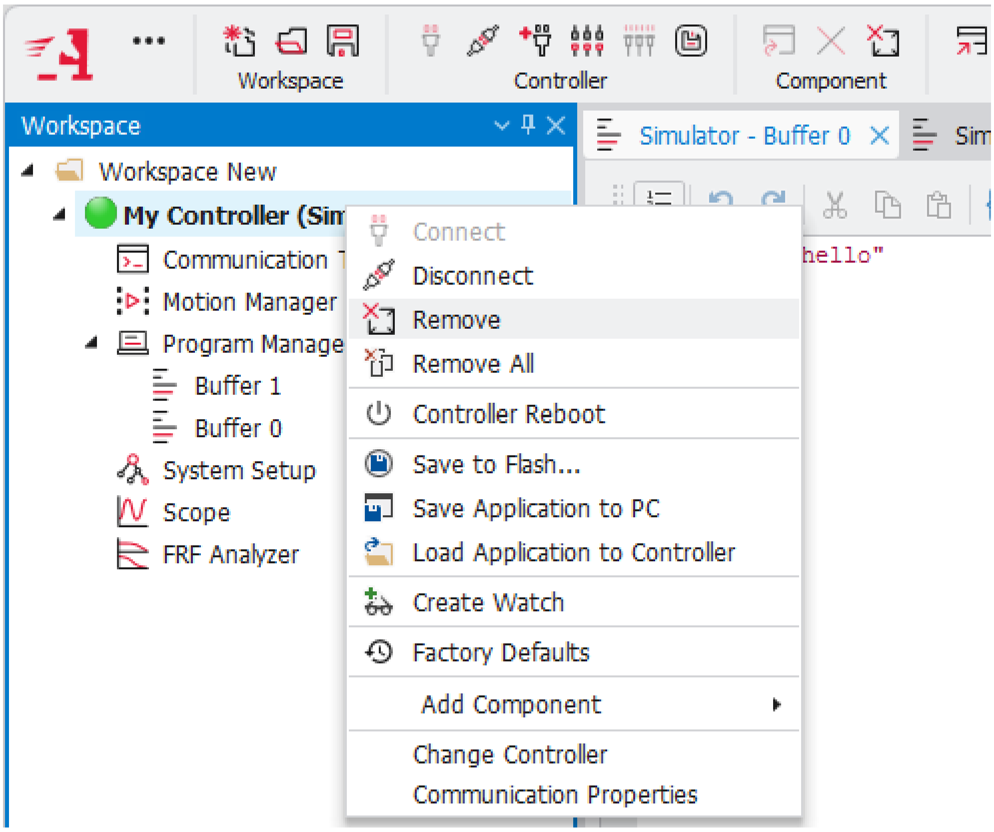

Figure 4-1. Remove My Controller (Simulator)

4. At the top of the window, select **Add Controller** (see Figure 4-2). This opens the **Connect to Controller** window (see Figure 4-3).

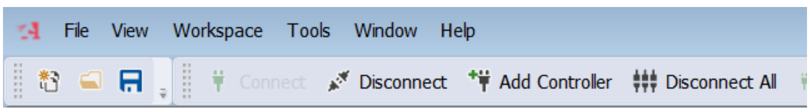

Figure 4-2. Add Controller

5. In the **Connect to Controller** window, click on **Ethernet** for the Connection Type. This displays the available Ethernet connections (see Figure 4-4).

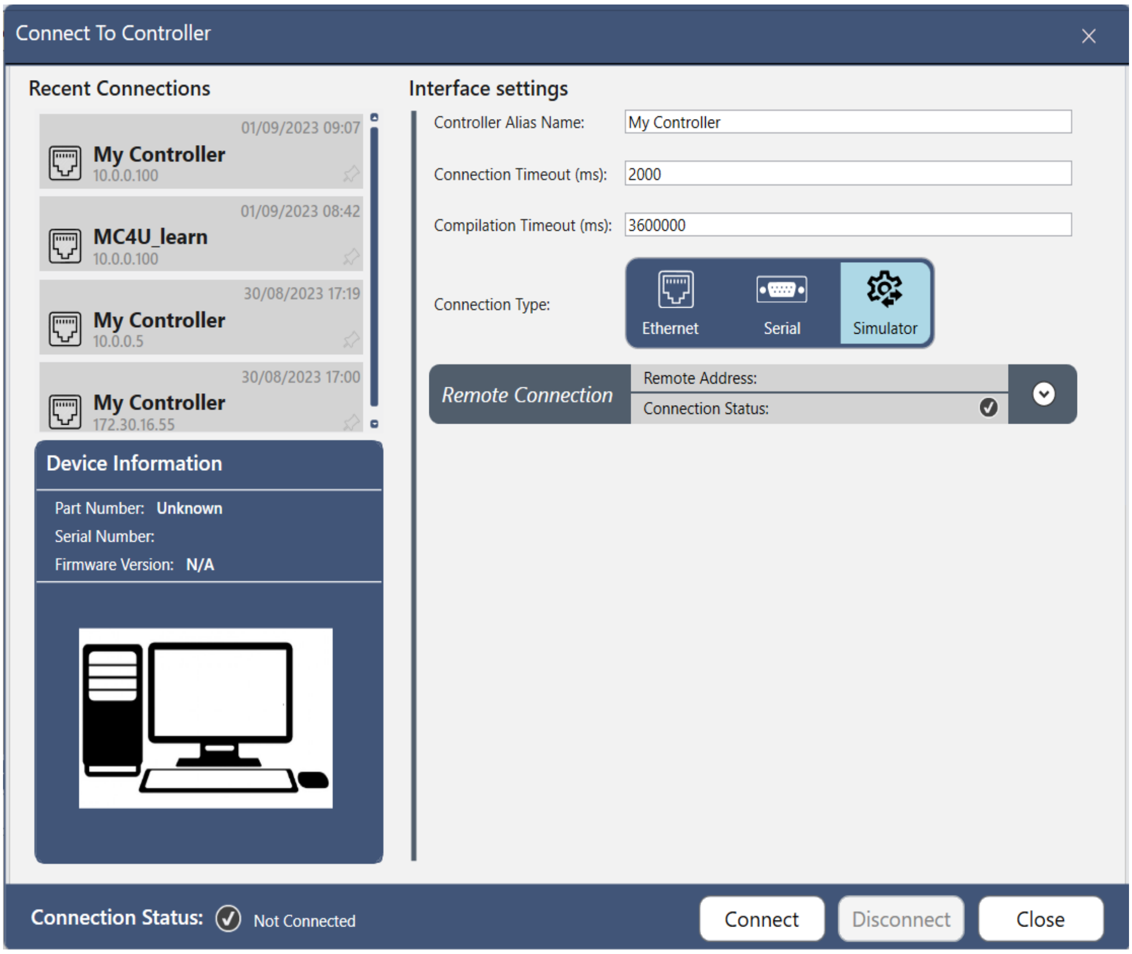

Figure 4-3. Connect to Controller Window

6. In the **Connect To Controller** window (Figure 4-4), select the Controller with the IP address 10.0.0.100 (the default IP address of the Controller). At the bottom of the window, click on **Connect**.
7. The connection status at the bottom left of the window shows a green check mark when the controller is connected to the MMI, as shown in Figure 4-4.

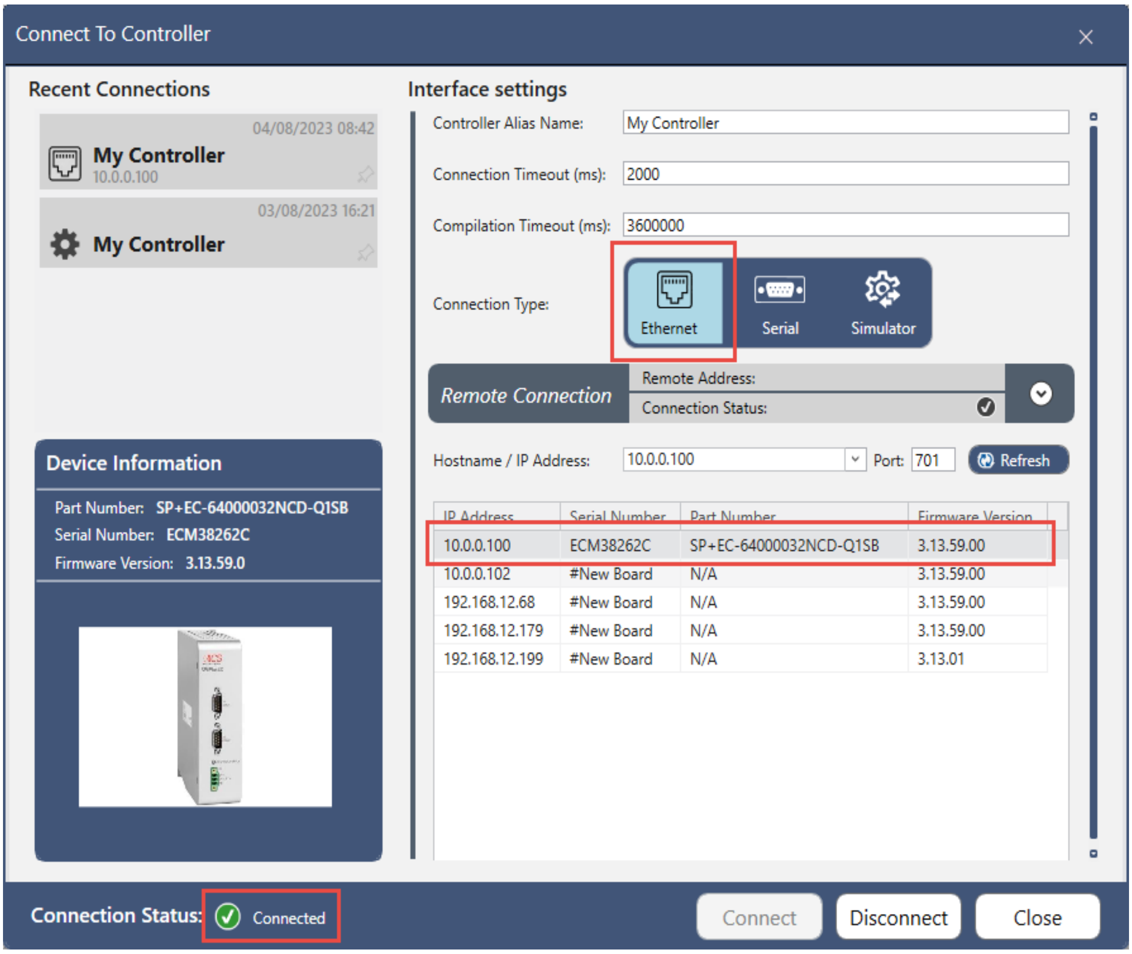

Figure 4-4. Select the Ethernet Connection

8. Close the **Connect to Controller** window.

5. Set Up the EtherCAT Network

This section IS NOT required for the MC4U and SPiiPlus CMxx that do not have a connected EtherCAT device.

The ECM series DOES NOT have EtherCAT functionality. If you are using an ECM device, skip this section.

To set up the EtherCAT network:

1. In the Workspace panel, right-click on the connected controller (see Figure 5-1), then navigate to **Add Component** > **Setup** > **System Setup**. This opens the **System Setup** window (see Figure 5-2).

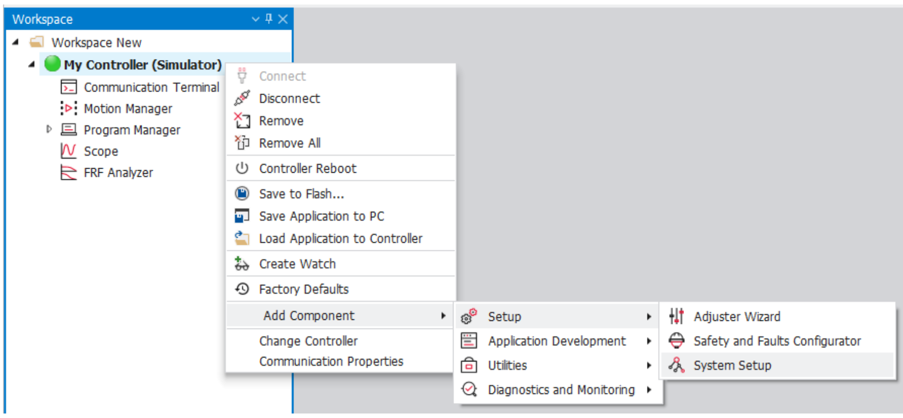

Figure 5-1. Right-Click on the Connected Controller

2. In the **Automatic Setup** tab, click **Execute** (Figure 5-2), and wait for the controller to scan the network and reboot. This may take several minutes.

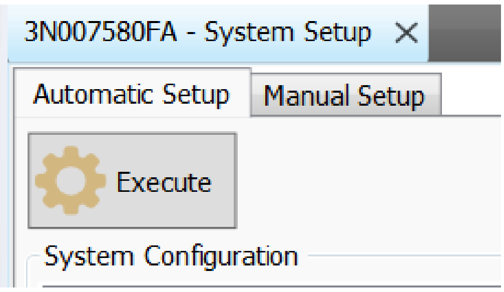

Figure 5-2. Execute Automatic Setup

3. Upon completion of the system setup, the **System Configuration** and **Product Information** panes are filled in (seeFigure 5-3).

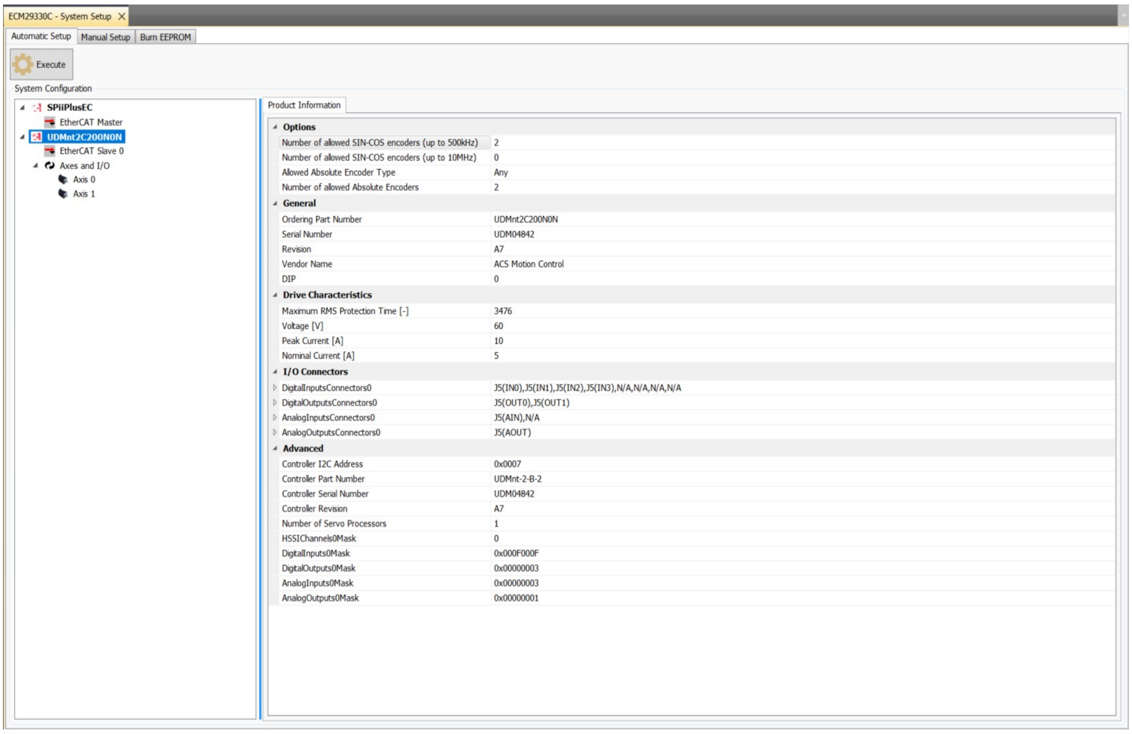

Figure 5-3. System Configuration After Automatic Setup

6. Use the Adjuster Wizard to Set up the Motors

Motor setup is handled through the Adjuster Wizard utility in the SPiiPlus MMI Studio. The previous sections must be completed before using the Adjuster Wizard.

The ACS website provides the following Tutorial Videos about the Adjuster Wizard:

- Tutorial 105 – Motor Setup and Tuning Part 1
- Tutorial 106 – Motor Setup & Tuning Part 2
- Tutorial 107 – Motor Setup and Tuning Part 3
- Tutorial 108 – Motor Setup and Tuning Part 4
- Tutorial 202 – Smarter Autotuning

Use this link to access these videos:

https://www.acsmotioncontrol.com/videos/tutorial-video-series/

To use the Adjuster Wizard to set up the motors:

1. In the Workspace panel of the SPiiPlus MMI Studio, right-click on the Controller, then select **Add Component** > **Setup** > **Adjusted Wizard** (see Figure 6-1).

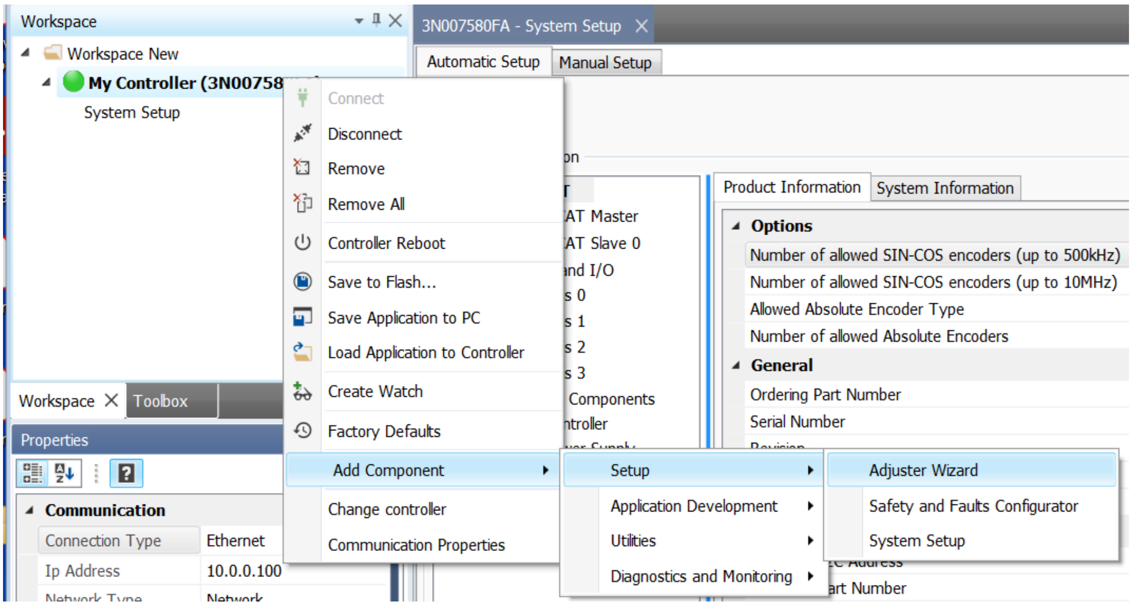

Figure 6-1. Select the Adjuster Wizard

2. From the first page of the **Adjuster Wizard**, you can proceed through the steps of setting up and exercising the motors. You select options on each page, then click on the **Next** button at the bottom of the page (see Figure 6-2).

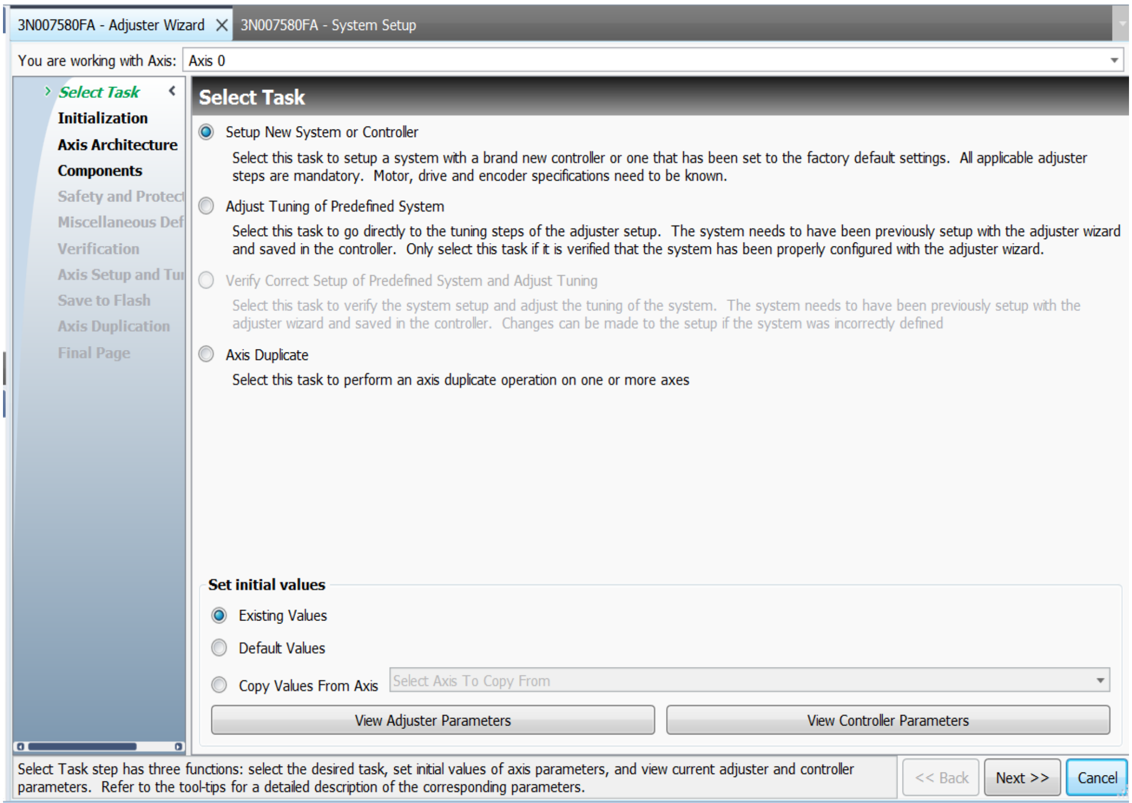

Figure 6-2. Adjuster Wizard: Select Task

3. Proceed through the **Adjuster Wizard** until you have completed motor setup.

7. Use the Motion Manager to Create Motion Profiles and Execute Moves

The Motion Manager tool allows you to create motion profiles and execute moves on axes that have previously been set up using the Adjuster Wizard.

To use the Motion Manager to create motion profiles and execute moves:

1. In the **Workspace**, right click on the **Controller**, then navigate to **Add Component** > **Diagnostics and Monitoring** > **Motion Manager**. This brings up the **Motion Manager** window (see Figure 7-1).

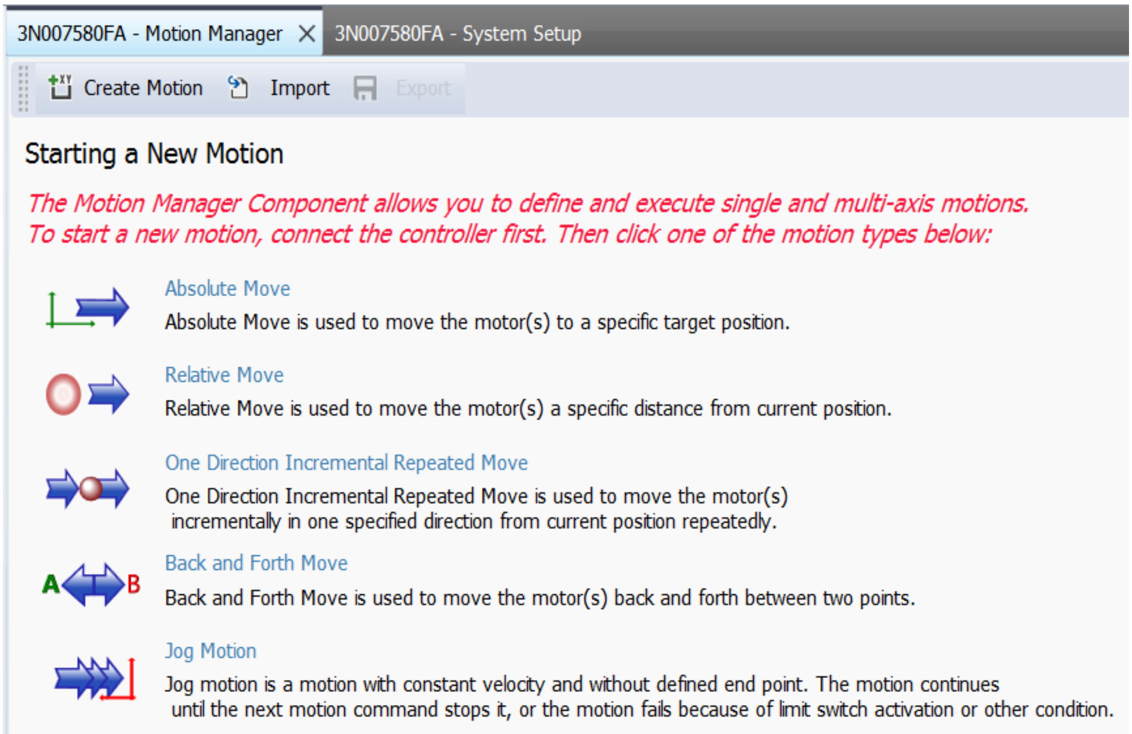

Figure 7-1. Motion Manager Initial Screen

2. To start, select the type of move you want from the list (see Figure 7-1).

> The simple moves shown on the first screen of the **Motion Manager** can verify that an axis is working as intended. More complex moves can be achieved by programming an application.

3. On the following screens, enter the desired motion parameters into the relevant fields.
4. Enable the selected motor and start the move. Verify that the motor moves as intended.

8. Scope

The SPiiPlus MMI Application Studio includes a Scope in which you can view a real-time graphic display of the motion. It provides a wide range of display options that allow you to define the graph's display according to your needs.

To open the **Scope**:

1. To open the **Scope**, right click on the **Controller**, then navigate to **Add Component** > **Diagnostics and Monitoring** > **Scope**. This opens the **Scope** screen (see Figure 8-1).
2. The **Settings** toolbar (see Figure 8-1) is collapsible. Expanding the **Settings** toolbar gives you access to parameters that expand the measurement capabilities of the **Scope**.

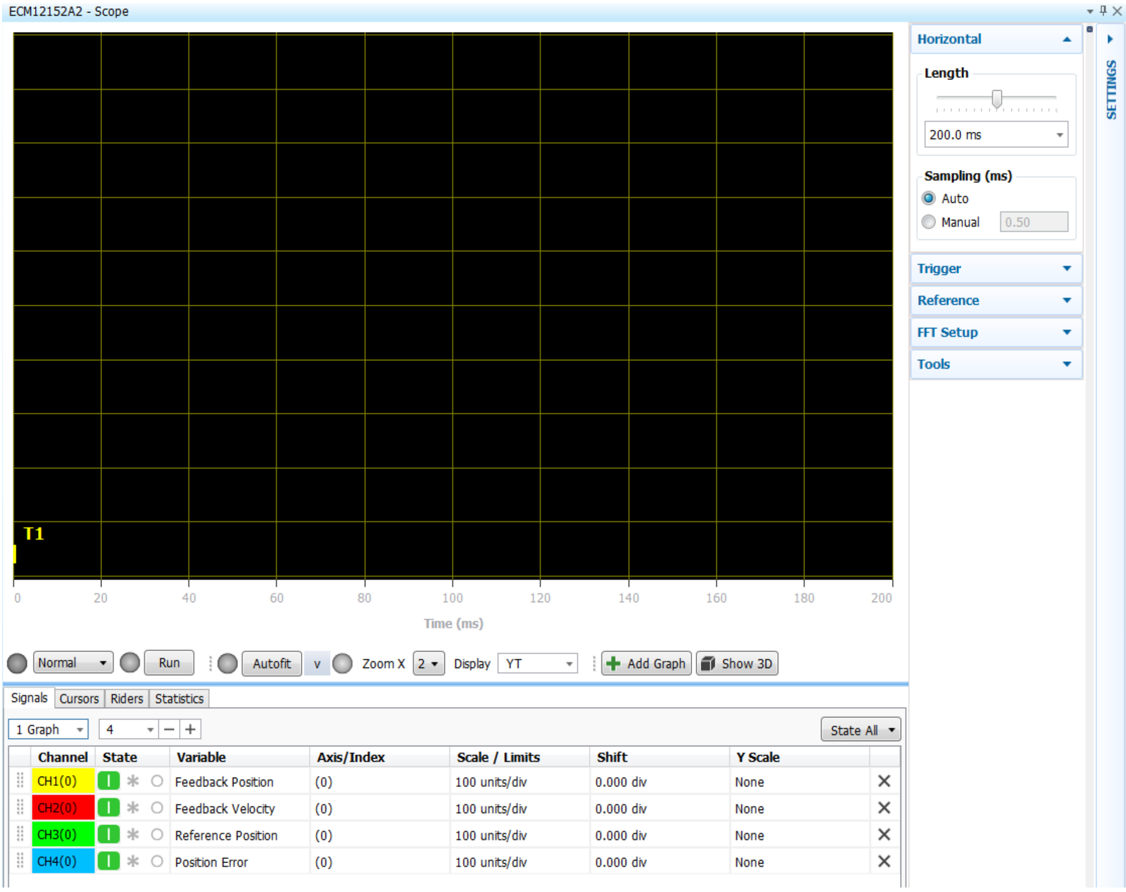

Figure 8-1. Scope

3. Figure 8-2 shows an example of some of the measurements that can be taken using the **Scope**. Note the following about Figure 8-2:
   - Select which parameters and axes are sampled using the **Signals** tab.
   - Change axes using the **Axes/Index** column.

> The **Scope** display can be triggered by selecting **Trigger Scope on Motion** checkbox in the **Motion Manager**.

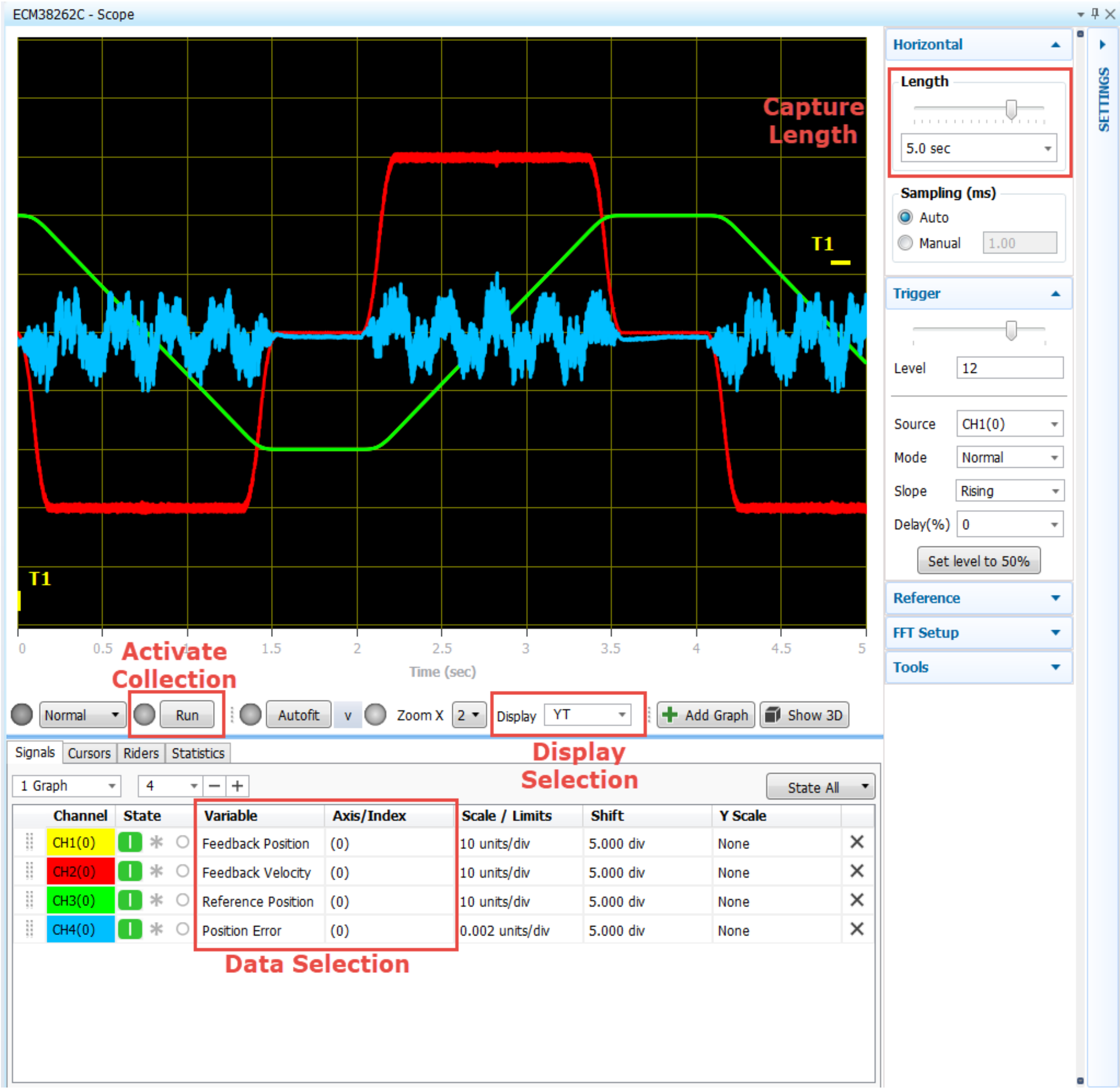

Figure 8-2. Example Display of the Scope

4. For more information about the **Scope**, see the MMI Application Studio User Guide.

9. More Information

ACS provides documentation and videos that help you get the most out of your ACS products. Following is a list of resources:

- SPiiPlus MMI Application Studio User Guide: downloaded onto the host computer along with the SPiiPlus Application Studio in:
  C:\Program Files (x86)\ACS Motion Control\SPiiPlus Documentation Kit/Software Guides.
- Programming in ACSPL+: ACSPL+ is a proprietary language developed by ACS for motion control applications. Resources for learning about ACSPL+ are available at:
  - ACS Tutorial video: Tutorial 201 – ACSPL+ Programming Introduction:
    https://www.acsmotioncontrol.com/resources/download/tutorial-201-acspl-programming-introduction/
  - ACS Technical Training videos, Sessions 3 and 4:
    https://www.acsmotioncontrol.com/videos/training-class-videos/
  - *ACSPL+ Programmers Guide* and its *Commands and Variables Reference Guide* were installed along with the SPiiPlus MMI Studio ADK in:
    C:\Program Files (x86)\ACS Motion Control\SPiiPlus Documentation Kit/Software Guides.
- Advanced Features and Algorithms: Topical Training Seminar videos are available on the ACS website through the following link:
  https://www.acsmotioncontrol.com/videos/topical-training-seminars/
  Subjects covered include:
  - Gantry configuration and tuning
  - Dual loop configuration and tuning
  - Stepper Motor open and closed loop configuration and tuning
  - Using IDM series drives with Beckhoff TwinCAT
  - Third-party EtherCAT device integration
  - Dynamic error compensation
  - Autotuning
  - Laser control with the LCI
  - Position capture and event generation with PEG and MARK

10. Contact ACS Support

ACS has Partners who sell and support our products. If you purchased our products from one of our Partners, contact them for support. Otherwise, select one of the options below.

10.1 Contact ACS Applications and Support

To contact support:

1. On the ACS website, click on **Technical Support** at the top of the page.
2. Fill out the blank fields on the **Submit a request** form.
3. Click **Submit** at the bottom of the form.

10.2 Submit a System Information Report

A System Information Report gives ACS information about your system that helps our applications engineers address issues you may have with your system.

To submit a system information report:

1. In the Workspace, right click on the Controller, then navigate to **Add Component** > **Utilities** > **Communication Terminal**. This opens the Communication Terminal at the bottom of the window.
2. In the field below the Serial Number of the Controller, enter `#SI`. Then click **Send** at the right side of the field (see Figure 10-1).
3. The Controller returns information about the system, which appears in the gray area below the **Send** button.
4. You can save the system information by clicking on **Save Printout** in the lower right of the Communication Terminal. Your system information is saved as a text file.
5. You can send the text file with the system information to an Application Engineer at ACS.

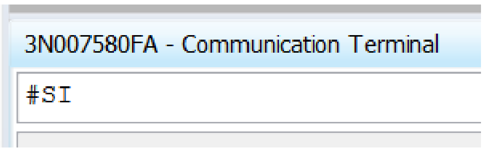

Figure 10-1. Communication Terminal